

Immersive Audio Signal Processing

Jens Blauert

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Immersive Audio Signal Processing

Sunil Bharitkar and Chris Kyriakakis

*Springer, 2006. 232 pp. Price: \$79.95 (hardcover).
ISBN: 038728532*

This book reports on recent and current research in audio-signal processing performed in Los Angeles at the Immersive Audio Laboratory, the Signal and Image Processing Institute (both at the University of Southern California) and the Audyssey Laboratories.

The main body of the book consists of six chapters, whereby five of them are marked as already published in journals or at conferences (Chaps. 4–7 and 9). These are preceded by three introductory chapters which deal with fundamentals of audio-signal processing; digital-filter design, acoustics, and psychoacoustics.

The focus of the book is clearly on digital-signal-processing methods, particularly for dealing with—almost exclusively linear—multichannel audio systems. The buzz word “immersive” in the book title denotes the fact that room-related systems for the creation of immersive auditory environments are usually multichannel ones. Yet, immersion as a perceptual effect—as currently heavily discussed among experts of virtual reality—is not dealt with.

The first of the main chapters (Chap. 4) starts with a virtual-microphone technique for creating surround sound, e.g., for deriving a plausible 5.1 audio perspective from two-channel stereo recordings. The possibility of parametric synthesis of musical sounds is discussed as a way out when virtual microphones in the traditional sense fail. Further, the well-known transaural technique for rendering binaural signals over loudspeakers is generalized for application with multiple listeners. Head tracking is not applied. The results of preliminary perceptual tests with realized systems are reported and a suggestion for further research is offered.

Chapter 5 elaborates on the theory of room equalization for multiple-listener positions in real time. The objective here is to design inverse filters for compensating the effects of loudspeakers and rooms. A pattern-recognition method is proposed to master the necessary simultaneous multichannel-equalization problem with low-order filters. Special emphasis is put on the evaluation of the robustness of the method to reverberation. A nonlinear projection method—the Sammon map, known from multidimensional scaling—is applied to visualize the results of the filter-design method used, i.e., a fuzzy clustering algorithm.

Chapter 6 regards practical problems of multichannel equalization, particularly, the selection of proper cross-over frequencies between loudspeakers. Specific topics are the equalization of phase differences between noncoincident loudspeakers and the design of multiposition bass-management filters. The very interesting results are evaluated by nonperceptual quality measures.

Chapter 7 concerns the effect of displacement on equalization, e.g., of listeners with regard to the measuring microphones. A mismatch analysis for spatial-average equalization is undertaken, resulting in observations of the

influence of distance and frequency on the equalization region, the steady-state equalization error depending on the listening distance and the influence of the reverberant-field correlation. For closer interpretation of these observations the reader is referred to future research.

Chapter 8 continues the topic of the preceding chapter, but from a different point of view. It reports on a simulation of a six-listener scenario using a statistical approach with modal equations for evaluating the robustness of equalization based on magnitude-response averaging. Perceptual verification of the results is suggested as a future project.

The last chapter (Chap. 9) takes on the problem of providing quiet zones within target spaces of audio systems. After a hardly convincing review of known methods to this end, the authors propose a novel approach based on eigenfilters. The properties of eigenfilters and their performance as a function of the filter order are discussed as well as performance sensitivity with respect to the lengths of the room responses. Perceptual tests are suggested as future work.

As most of the main body of the book has already been peer-reviewed in the process of prior publications, there is no reason for this reviewer to question the soundness of the results. The main part of the book can thus be seen as a proceedings report from the authors' laboratories.

The introductory part of the book starts with two chapters on fundamentals of digital signal processing. Although clearly written, they represent a syllabus rather than an introduction for readers not yet familiar with the basics of the field.

The third chapter regards fundamentals of acoustics and psychoacoustics. Here the suitability for teaching purposes must be seriously questioned. As to the acoustics, even if one agreed with the selection of topics, the presentation is inadequate. For example, many equations are written in such a way that letter symbols denote numbers instead of quantities. At some point even the good old “erg” celebrates a revival!

Regarding the psychoacoustics part, it is obvious that the authors are no experts in this field. Their explanation of the frequency selectivity in the cochlea is adventurous. This reviewer asked himself why the authors have included a psychoacoustics section at all, since this field is not actually applied in the main body of the book.

In the foreword, the expectation is expressed that this little book may be useful as a text for courses in digital audio-signal processing. This may well be the case, but prospective users are warned that they may need additional course material for this purpose.

The book, however, provides a compact overview on an important research area within digital audio-signal processing at large. For those who want to avoid the effort of collecting the original articles, this compilation is certainly a good buy.

JENS BLAUERT
Communications Acoustics
Ruhr University
Universtaetsstr 150
Bochum, D-44780 Germany